Question

Partial electrode potentials are shown below:

Half-reaction	Standard Electrode Potential
$4\mathbf{A}O_2(aq) + 4H^+ + 6e^- \rightarrow \mathbf{A}_4O_6^{2-} + 2H_2O$	0.51V
$2\mathbf{A}O_2(aq) + 2H^+ + 4e^- \rightarrow \mathbf{A}_2O_3^{2-} + H_2O$	0.40V
$*2\mathbf{A}O_3^{2-} + 3H_2O + 4e^- \rightarrow \mathbf{A}_2O_3^{2-} + 6OH^-$	-0.58V
$*\mathbf{A}O_4^{2-} + 4H_2O + 6e^- \rightarrow \mathbf{A} + 8OH^-$	-0.66V
$\mathbf{B}_2 + 2e^- \rightarrow 2\mathbf{B}^-$	0.535V

Note: The standard electrode potentials of reactions marked with "$*$" are values at $\mathrm{pH{=}14}$; in this problem, the temperature is assumed to be 298K, and the two-stage ionization constants of $H_2\mathbf{A}O_3$ are 1.3×10^{-2} and 6.3×10^{-8}, respectively.

Then, the equilibrium constant K_1 for the titration of $\mathbf{B}_2$ by $\mathbf{A}_2O_3^{2-}$ (one of the products is $\mathbf{A}_4O_6^{2-}$) and the equilibrium constant K_2 for the reaction $\mathbf{A}O_2 + H_2O \rightarrow H_2\mathbf{A}O_3$ are respectively (retain two significant figures, an error within 5% is considered correct):

A. All other options are incorrect

B. 5.3×10^{13}, 1.7

C. 5.3×10^{13}, 1.5

D. 5.3×10^{14}, 1.7

E. 5.3×10^{14}, 1.5

F. 5.3×10^{15}, 1.7

G. 5.3×10^{15}, 1.5

H. 3.5×10^{13}, 1.7

I. 3.5×10^{13}, 1.5

J. 3.5×10^{14}, 1.7

K. 3.5×10^{14}, 1.5

L. 3.5×10^{15}, 1.7

M. 3.5×10^{15}, 1.5

N. 5.3×10^{13}, 1.9

O. 5.3×10^{14}, 1.9

P. 5.3×10^{15}, 1.9

Q. 3.5×10^{13}, 1.9

R. 3.5×10^{14}, 1.9

S. 3.5×10^{15}, 1.9

Answer

Correct Answer: F

Detailed Explanation

Calculate K_1

Connect the battery in series with a standard hydrogen electrode, and calculate the change in Gibbs free energy of the reaction (hydrogen electrode changes are omitted here)

$$4\mathbf{A}O_2(aq) + 4H^+ + 8e^- \rightarrow 2\mathbf{A}_2O_3^{2-} + 2H_2O$$

$$\Delta_r G_m^\circ = -nEF = -308.752kJ/mol$$

$$4\mathbf{A}O_2(aq) + 4H^+ + 6e^- \rightarrow \mathbf{A}_4O_6^{2-} + 2H_2O$$

$$\Delta_r G_m^\circ = -nEF = -295.244kJ/mol$$

$$\mathbf{A}_4O_6^{2-} + 2e^- \rightarrow 2\mathbf{A}_2O_3^{2-}$$

$$\Delta_r G_m^\circ = -13.508kJ/mol$$

Therefore $E = 0.070V$

CHECKPOINT 2 PTS

The standard electrode potential of $\mathbf{A}_4O_6^{2-} + 2e^- \rightarrow 2\mathbf{A}_2O_3^{2-}$ is 0.070V

Therefore, the electromotive force of $\mathbf{A}_2O_3^{2-}$ titrating $\mathbf{B}_2$ is 0.465V,

$$K_1 = e^{\frac{nEF}{RT}} = 5.3 \times 10^{15}$$

CHECKPOINT 1 PTS

The equilibrium constant K_1 is 5.3×10^{15}

Calculate K_2

Convert the data under alkaline conditions:

$$2\mathbf{A}O_3^{2-} + 6H^+ + 4e^- \rightarrow \mathbf{A}_2O_3^{2-} + 3H_2O$$

$$E' = 0.6623V$$

$$2\mathbf{A}O_3^{2-} + 2H^+ + 2H_2 \rightarrow \mathbf{A}_2O_3^{2-} + 3H_2O$$

$$K = 6.051 \times 10^{44}$$

$$2\mathbf{A}O_2 + 2H_2 \rightarrow \mathbf{A}_2O_3^{2-} + H_2O + 2H^+$$

$$K = 1.114 \times 10^{27}$$

Subtract to get

$$\mathbf{A}O_2 + H_2O \rightarrow \mathbf{A}O_3^{2-} + 2H^+$$

$$K = 1.357 \times 10^{-9}$$

CHECKPOINT 1 PTS

The equilibrium constant of $\mathbf{A}O_2 + H_2O \rightarrow \mathbf{A}O_3^{2-} + 2H^+$ is $K = 1.357 \times 10^{-9}$

Thus we obtain

$$\mathbf{A}O_2 + H_2O \to H_2\mathbf{A}O_3$$

$$K = 1.656$$

CHECKPOINT 2 PTS

The equilibrium constant of $\mathbf{A}O_2 + H_2O \to H_2\mathbf{A}O_3$ is $K_2 = 1.656$